APPENDIX

In this appendix, we provide the detailed algebraic derivations supporting the stability claims made in Theorem 1. By constructing a comprehensive Lyapunov function candidate, we establish the uniform boundedness of the neural network weight estimation errors and prove the finite-time stability of the state estimation error for the 4WID electric vehicle system.

A. Convergence Analysis of the Error Dynamics

Define the state and weight estimation errors as

$$\begin{aligned} \boldsymbol{e}_\zeta &= \boldsymbol{\zeta} - \hat{\boldsymbol{\zeta}}, \\ \widetilde{\boldsymbol{W}}_g &= \boldsymbol{W}_g^* - \hat{\boldsymbol{W}}_g, \\ \widetilde{\boldsymbol{W}}_\xi &= \boldsymbol{W}_\xi^* - \hat{\boldsymbol{W}}_\xi, \end{aligned} \tag{A.1}$$

where $\boldsymbol{W}_g^*$ and $\boldsymbol{W}_\xi^*$ are the constant ideal weights in the RBF approximations. For brevity, write $\boldsymbol{H}_g = \boldsymbol{H}_g(\boldsymbol{X}_g)$ and $\boldsymbol{H}_\xi = \boldsymbol{H}_\xi(\boldsymbol{X}_\xi)$. We use the adaptive laws in the main text, with $\boldsymbol{S} = [\boldsymbol{I}_3^\top, \boldsymbol{0}_{3\times 4}]^\top$ and leakage gains $k_1, k_2 > 0$.

Let $\boldsymbol{d}_\omega(t)$ collect the four additive fault terms $k_i \Delta u_i / I_\omega$ in the order $\mathrm{fl}, \mathrm{fr}, \mathrm{rl}, \mathrm{rr}$. Subtracting the observer equation from the augmented plant equation and applying the RBF approximations gives

$$\begin{aligned} \dot{\boldsymbol{e}}_\zeta = \Delta \boldsymbol{F}_3 &+ \operatorname{diag}(\boldsymbol{U}_1)\widetilde{\boldsymbol{W}}_g^\top \boldsymbol{H}_g \\ &+ \boldsymbol{S}\widetilde{\boldsymbol{W}}_\xi^\top \boldsymbol{H}_\xi - \boldsymbol{L}_f \boldsymbol{\psi}_n(\boldsymbol{e}_\zeta) + \boldsymbol{r}_\zeta(t), \end{aligned} \tag{A.2}$$

where

$$\begin{aligned} \Delta \boldsymbol{F}_3 &= \boldsymbol{F}_3(\boldsymbol{\zeta}; \boldsymbol{x}, \delta) - \boldsymbol{F}_3(\hat{\boldsymbol{\zeta}}; \boldsymbol{x}, \delta), \\ \boldsymbol{r}_\zeta(t) &= \operatorname{diag}(\boldsymbol{U}_1)\boldsymbol{\varepsilon}_g(t) + \begin{bmatrix} \boldsymbol{\varepsilon}_\xi(t) \\ \boldsymbol{d}_\omega(t) \end{bmatrix}, \end{aligned} \tag{A.3}$$

and $\boldsymbol{\psi}_n(\boldsymbol{e}_\zeta)$ has components $\operatorname{sgn}(e_{\zeta,j})|e_{\zeta,j}|^{n-1}$, with $1 < n < 2$. Both evaluations of $\boldsymbol{F}_3$ use the same measured vehicle states $\boldsymbol{x}$ and steering angle δ as external inputs.

Assume that $\|\boldsymbol{U}_1(t)\|_\infty \le \bar{U}_1$ and $|\Delta u_i(t)| \le \overline{\Delta u}_i$. The RBF approximation error bounds then yield

$$\begin{aligned} &\|\boldsymbol{r}_\zeta(t)\|_2 \le \bar{r}_\zeta, \\ &\bar{r}_\zeta = \bar{U}_1 \bar{\varepsilon}_g + \left(\bar{\varepsilon}_\xi^2 + \sum_{i\in\{\mathrm{fl,fr,rl,rr}\}} \frac{k_i^2 \overline{\Delta u}_i^{\,2}}{I_\omega^2} \right)^{1/2}. \end{aligned} \tag{A.4}$$

Choose the Lyapunov function

$$V_1 = \boldsymbol{e}_\zeta^\top \boldsymbol{P} \boldsymbol{e}_\zeta + \operatorname{tr}\!\left(\widetilde{\boldsymbol{W}}_g^\top \boldsymbol{Q} \widetilde{\boldsymbol{W}}_g\right) + \operatorname{tr}\!\left(\widetilde{\boldsymbol{W}}_\xi^\top \boldsymbol{R} \widetilde{\boldsymbol{W}}_\xi\right), \tag{A.5}$$

where $\boldsymbol{Q}$ and $\boldsymbol{R}$ are symmetric positive definite. For the following sufficient conditions, take $\boldsymbol{P} = \operatorname{diag}(p_1, \ldots, p_7)$ and $\boldsymbol{L}_f = \operatorname{diag}(\ell_1, \ldots, \ell_7)$ with $p_j, \ell_j > 0$. The diagonal entries p_j here refer to $\boldsymbol{P}$.

Since the ideal weights are constant, $\dot{\widetilde{\boldsymbol{W}}}_g = -\dot{\hat{\boldsymbol{W}}}_g$ and $\dot{\widetilde{\boldsymbol{W}}}_\xi = -\dot{\hat{\boldsymbol{W}}}_\xi$. Differentiating V_1 gives

$$\dot{V}_1 = 2\boldsymbol{e}_\zeta^\top \boldsymbol{P} \dot{\boldsymbol{e}}_\zeta - 2\operatorname{tr}\!\left(\widetilde{\boldsymbol{W}}_g^\top \boldsymbol{Q} \dot{\hat{\boldsymbol{W}}}_g\right) - 2\operatorname{tr}\!\left(\widetilde{\boldsymbol{W}}_\xi^\top \boldsymbol{R} \dot{\hat{\boldsymbol{W}}}_\xi\right). \tag{A.6}$$

The weight cross terms cancel upon substitution of the adaptive laws. For the gain approximation, this follows from

$$\begin{aligned} &\boldsymbol{e}_\zeta^\top \boldsymbol{P} \operatorname{diag}(\boldsymbol{U}_1)\widetilde{\boldsymbol{W}}_g^\top \boldsymbol{H}_g \\ &\quad = \operatorname{tr}\!\left(\widetilde{\boldsymbol{W}}_g^\top \boldsymbol{H}_g \boldsymbol{e}_\zeta^\top \boldsymbol{P} \operatorname{diag}(\boldsymbol{U}_1)\right). \end{aligned} \tag{A.7}$$

The disturbance approximation term cancels by the same trace identity with $\boldsymbol{S}$ in place of $\operatorname{diag}(\boldsymbol{U}_1)$. Consequently,

$$\begin{aligned} \dot{V}_1 = &\, 2\boldsymbol{e}_\zeta^\top \boldsymbol{P} \Delta \boldsymbol{F}_3 - 2\boldsymbol{e}_\zeta^\top \boldsymbol{P} \boldsymbol{L}_f \boldsymbol{\psi}_n(\boldsymbol{e}_\zeta) \\ &+ 2\boldsymbol{e}_\zeta^\top \boldsymbol{P} \boldsymbol{r}_\zeta + 2k_1 \operatorname{tr}\!\left(\widetilde{\boldsymbol{W}}_g^\top \hat{\boldsymbol{W}}_g\right) + 2k_2 \operatorname{tr}\!\left(\widetilde{\boldsymbol{W}}_\xi^\top \hat{\boldsymbol{W}}_\xi\right). \end{aligned} \tag{A.8}$$

For either weight matrix, the leakage term satisfies

$$\begin{aligned} 2k \operatorname{tr}\!\left(\widetilde{\boldsymbol{W}}^\top \hat{\boldsymbol{W}}\right) &= k\|\boldsymbol{W}^*\|_F^2 - k\|\widetilde{\boldsymbol{W}}\|_F^2 - k\|\hat{\boldsymbol{W}}\|_F^2 \\ &\le -k\|\widetilde{\boldsymbol{W}}\|_F^2 + k\|\boldsymbol{W}^*\|_F^2. \end{aligned} \tag{A.9}$$

Fix $M > 0$ and consider $V_1 \le M$. Assume that the true states and the estimates in this sublevel set lie in a bounded operating domain on which the longitudinal-speed and slip-ratio denominators have positive lower bounds. Assume also that $\|\Delta \boldsymbol{F}_3\|_2 \le K_e \|\boldsymbol{e}_\zeta\|_2$ uniformly on this domain. Define

$$E_M = \sqrt{\frac{M}{\lambda_{\min}(\boldsymbol{P})}}, \qquad \ell_* = \min_{1\le j\le 7} p_j \ell_j. \tag{A.10}$$

On $V_1 \le M$, we have $\|\boldsymbol{e}_\zeta\|_2 \le E_M$ and

$$\begin{aligned} \boldsymbol{e}_\zeta^\top \boldsymbol{P} \boldsymbol{L}_f \boldsymbol{\psi}_n(\boldsymbol{e}_\zeta) &= \sum_{j=1}^{7} p_j \ell_j |e_{\zeta,j}|^n \ge \ell_* \|\boldsymbol{e}_\zeta\|_2^n, \\ \boldsymbol{e}_\zeta^\top \boldsymbol{P} \Delta \boldsymbol{F}_3 &\le \|\boldsymbol{P}\|_2 K_e E_M^{2-n} \|\boldsymbol{e}_\zeta\|_2^n. \end{aligned} \tag{A.11}$$

Impose the gain condition

$$d_M = \ell_* - \|\boldsymbol{P}\|_2 K_e E_M^{2-n} > 0. \tag{A.12}$$

For $b_\zeta = 2\|\boldsymbol{P}\|_2 \bar{r}_\zeta$, Young's inequality gives

$$\begin{aligned} b_\zeta \|\boldsymbol{e}_\zeta\|_2 &\le d_M \|\boldsymbol{e}_\zeta\|_2^n + C_r, \\ C_r &= (n-1) d_M \left(\frac{b_\zeta}{n d_M}\right)^{n/(n-1)}. \end{aligned} \tag{A.13}$$

It follows from (A.8) that

$$\begin{aligned} \dot{V}_1 \le &\, -d_M \|\boldsymbol{e}_\zeta\|_2^n - k_1 \|\widetilde{\boldsymbol{W}}_g\|_F^2 - k_2 \|\widetilde{\boldsymbol{W}}_\xi\|_F^2 \\ &+ C_r + k_1 \|\boldsymbol{W}_g^*\|_F^2 + k_2 \|\boldsymbol{W}_\xi^*\|_F^2. \end{aligned} \tag{A.14}$$

Let $p = n/2 \in (0, 1)$, and let X and Y denote the state term and the sum of the two weight terms in V_1, respectively. Define

$$\gamma = \min\left\{ \frac{d_M}{\lambda_{\max}(\boldsymbol{P})^p}, \frac{k_1}{\lambda_{\max}(\boldsymbol{Q})}, \frac{k_2}{\lambda_{\max}(\boldsymbol{R})} \right\} > 0. \tag{A.15}$$

The preceding bound implies

$$\begin{aligned} \dot{V}_1 &\le -\gamma(X^p + Y) + C_r + k_1 \|\boldsymbol{W}_g^*\|_F^2 + k_2 \|\boldsymbol{W}_\xi^*\|_F^2 \\ &\le -\gamma V_1^p + \eta, \end{aligned} \tag{A.16}$$

where the last step uses $Y^p \le Y+1$ and $(X+Y)^p \le X^p+Y^p$, and

$$\eta = C_r + k_1\|\boldsymbol{W}_g^*\|_F^2 + k_2\|\boldsymbol{W}_\xi^*\|_F^2 + \gamma. \tag{A.17}$$

Suppose $V_1(0) \le M$ and $\gamma M^p > \eta$. Then $\dot V_1 < 0$ on $V_1 = M$, so the sublevel set $V_1 \le M$ is forward invariant. Choose $\vartheta \in (0,1)$ such that

$$V_* = \left(\frac{\eta}{\gamma(1-\vartheta)}\right)^{1/p} < M. \tag{A.18}$$

Applying the practical finite-time stability lemma in the main text on the invariant set $V_1 \le M$ gives the entry time

$$T \le \frac{[V_1(0)^{1-p} - V_*^{1-p}]_+}{\gamma\vartheta(1-p)}, \qquad [s]_+ = \max\{s,0\}. \tag{A.19}$$

The trajectory remains in $V_1 \le V_*$ for all $t \ge T$, so

$$\begin{aligned} \|\boldsymbol{e}_\zeta(t)\|_2 &\le \sqrt{V_*/\lambda_{\min}(\boldsymbol{P})}, \\ \|\widetilde{\boldsymbol{W}}_g(t)\|_F &\le \sqrt{V_*/\lambda_{\min}(\boldsymbol{Q})}, \\ \|\widetilde{\boldsymbol{W}}_\xi(t)\|_F &\le \sqrt{V_*/\lambda_{\min}(\boldsymbol{R})}. \end{aligned} \tag{A.20}$$

Thus, the state and weight estimation errors are uniformly bounded and enter the stated neighborhoods in finite time under the sufficient conditions above.

B. Stability Analysis of the Controlled System

Substituting the nominal control law into the transformed error dynamics gives

$$\begin{aligned} \dot{\boldsymbol{z}} &= -\boldsymbol{\kappa}\boldsymbol{z} + \boldsymbol{w}_z(t), \\ \boldsymbol{w}_z(t) &= \boldsymbol{D}\big[\boldsymbol{\xi}_{\mathrm{c}}(t) - \hat{\boldsymbol{\xi}}_{\mathrm{c}}(t) + \boldsymbol{B}_3(\boldsymbol{U}-\boldsymbol{U}_s)\big]. \end{aligned} \tag{A.21}$$

Under the conditions of the closed-loop theorem, $|z_i(0)| < 1$ and $|w_{z,i}(t)| \le \bar w_i < \kappa_i$ for $i = 1,\dots,4$. Variation of constants gives

$$z_i(t) = e^{-\kappa_i t} z_i(0) + \int_0^t e^{-\kappa_i(t-\tau)} w_{z,i}(\tau)\, d\tau. \tag{A.22}$$

Consequently,

$$\begin{aligned} |z_i(t)| &\le e^{-\kappa_i t}|z_i(0)| + \frac{\bar w_i}{\kappa_i}(1-e^{-\kappa_i t}) \\ &\le \rho_i, \qquad \rho_i = \max\left\{|z_i(0)|, \frac{\bar w_i}{\kappa_i}\right\} < 1. \end{aligned} \tag{A.23}$$

Thus $|e_i(t)| < p_i(t)$ for all $t \ge 0$, and

$$\begin{aligned} 0 \le V_2(t) &= \frac{k_0}{2}\sum_{i=1}^4 \log\left(\frac{1}{1-z_i^2(t)}\right) \\ &\le \frac{k_0}{2}\sum_{i=1}^4 \log\left(\frac{1}{1-\rho_i^2}\right) < \infty. \end{aligned} \tag{A.24}$$

If $\boldsymbol{w}_z(t)$ vanishes asymptotically, then for any $\epsilon > 0$ there exists T_ϵ such that $|w_{z,i}(t)| \le \kappa_i\epsilon$ for $t \ge T_\epsilon$. Applying variation of constants from T_ϵ yields

$$\begin{aligned} |z_i(t)| &\le e^{-\kappa_i(t-T_\epsilon)}|z_i(T_\epsilon)| \\ &\quad + \epsilon\big(1 - e^{-\kappa_i(t-T_\epsilon)}\big). \end{aligned} \tag{A.25}$$

Hence $\limsup_{t\to\infty}|z_i(t)| \le \epsilon$. Since ϵ is arbitrary, $\boldsymbol{z}(t)$ converges asymptotically to zero. Continuity of V_2 at the origin then implies that $V_2(t)$ converges to zero.

C. Stability Analysis of the Controlled System

The transformed error dynamics in the main text are

$$\dot{\boldsymbol{z}} = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{D}\boldsymbol{F}_4 + \boldsymbol{D}\boldsymbol{B}_3\boldsymbol{U} + \boldsymbol{D}\boldsymbol{\xi}_{\mathrm{c}}. \tag{A.26}$$

The nominal control input satisfies

$$\boldsymbol{D}\boldsymbol{B}_3\boldsymbol{U}_s = -\boldsymbol{A}\boldsymbol{z} - \boldsymbol{D}\boldsymbol{F}_4 - \boldsymbol{D}\hat{\boldsymbol{\xi}}_{\mathrm{c}} - \boldsymbol{\kappa}\boldsymbol{z}. \tag{A.27}$$

Writing $\boldsymbol{U} = \boldsymbol{U}_s + (\boldsymbol{U}-\boldsymbol{U}_s)$ and substituting the nominal control relation yields

$$\begin{aligned} \dot{\boldsymbol{z}} &= -\boldsymbol{\kappa}\boldsymbol{z} + \boldsymbol{w}_z(t), \\ \boldsymbol{w}_z(t) &= \boldsymbol{D}\big[\boldsymbol{\xi}_{\mathrm{c}}(t) - \hat{\boldsymbol{\xi}}_{\mathrm{c}}(t) + \boldsymbol{B}_3(\boldsymbol{U}-\boldsymbol{U}_s)\big]. \end{aligned} \tag{A.28}$$

Suppose $|z_i(0)| < 1$ and $|w_{z,i}(t)| \le \bar w_i < \kappa_i$ for $i = 1,\dots,4$, as required by the closed-loop theorem. Since $\boldsymbol{\kappa}$ is diagonal, variation of constants gives

$$z_i(t) = e^{-\kappa_i t} z_i(0) + \int_0^t e^{-\kappa_i(t-\tau)} w_{z,i}(\tau)\, d\tau. \tag{A.29}$$

Taking absolute values yields

$$\begin{aligned} |z_i(t)| &\le e^{-\kappa_i t}|z_i(0)| + \frac{\bar w_i}{\kappa_i}(1-e^{-\kappa_i t}) \\ &\le \rho_i, \qquad \rho_i = \max\left\{|z_i(0)|, \frac{\bar w_i}{\kappa_i}\right\} < 1. \end{aligned} \tag{A.30}$$

Therefore, $|e_i(t)| < p_i(t)$ for all $t \ge 0$. The barrier Lyapunov function satisfies

$$\begin{aligned} 0 \le V_2(t) &= \frac{k_0}{2}\sum_{i=1}^4 \log\left(\frac{1}{1-z_i^2(t)}\right) \\ &\le \frac{k_0}{2}\sum_{i=1}^4 \log\left(\frac{1}{1-\rho_i^2}\right) < \infty. \end{aligned} \tag{A.31}$$

If $\boldsymbol{w}_z(t)$ vanishes asymptotically, then for any $\epsilon > 0$ there exists a time T_ϵ such that $|w_{z,i}(t)| \le \kappa_i\epsilon$ for all $t \ge T_\epsilon$ and $i = 1,\dots,4$. Applying variation of constants from T_ϵ gives

$$\begin{aligned} |z_i(t)| &\le e^{-\kappa_i(t-T_\epsilon)}|z_i(T_\epsilon)| \\ &\quad + \epsilon\big(1 - e^{-\kappa_i(t-T_\epsilon)}\big). \end{aligned} \tag{A.32}$$

Hence $\limsup_{t\to\infty}|z_i(t)| \le \epsilon$. Since ϵ is arbitrary, each $z_i(t)$ converges asymptotically to zero. Continuity of V_2 at $\boldsymbol{z} = \boldsymbol{0}$ then implies that $V_2(t)$ also converges to zero.